2024 IEEE International Conference on Image Processing (ICIP) | 979-8-3503-4939-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICIP51287.2024.10647498

# LEGIT: TEXT LEGIBILITY FOR USER-GENERATED MEDIA

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## ABSTRACT

User-generated content (UGC) is ubiquitous across the internet as a result of billions of videos and images being uploaded each day. All kinds of UGC media are affected by natural distortions, occurring both during and after capture, which are inherently diverse and commingled. These distortions have different perceptual effects based on the media content. Given recent dramatic increases in the consumption of short-form content, the analysis and control of their perceptual quality has become an important problem. Regardless of the content, many UGC videos have overlaid and embedded texts in them, which are visually salient. Hence text quality has a significant impact on the global perception of video or image quality and needs to be studied. One of the most important factors in perceptual text quality in user-generated media is legibility, which has been studied very little in the context of computer vision. Predicting text legibility can also help in text recognition applications such as image search or document identification. This work aims at modeling text legibility using computer vision techniques and thus studying the relationship between text quality and legibility. We propose a modified dataset variant of COCO-Text [1] and a model for predicting text legibility for both handwritten and machine-generated texts. We also demonstrate how models trained to predict text legibility can help in the prediction of text (perceptual) quality. The dataset and models can be accessed here https://live.ece.utexas.edu/research/Quality/index.htm.

***Index Terms*—** Image Quality Assessment, Text Quality, Text Legibility, User-generated Content

## 1. INTRODUCTION

UGC is one of the most popular forms of media consumption for billions of users across several platforms like Instagram, YouTube, and TikTok [2]. Among the different forms of UGC, short-form content has over 1.6 billion users and is projected to represent 80% of all mobile data traffic in North America, becoming the most popular type of content on social media [3]. Given the popularity of short-form UGC, it is important to study and control the different aspects of this kind of media and how its content can affect the perceptual quality. Predicting and controlling the perceptual quality of UGC videos can increase user with engagement thereby improving the ROI of businesses operating popular media platforms.

**Fig. 1**: **Creating the modified COCO-Text Datatset:** The small text regions (left picture) labeled in the COCO-Text [1] dataset were merged to create text patches (shown in green bounding boxes on right) used for training the legibility models. Best viewed in color.

Textual content, being highly salient [4], affects the perception of visual media, and the prevalence of overlaid and embedded texts in short-form UGC generates the need to study the quality issues associated with them. Text quality can be affected by the capture process, the generator application, and the post-processing on the platforms they are uploaded to. Given the diversity of textual content - machine-generated or handwritten, embedded or overlaid, language and style variations, etc. makes the problem of determining the legibility of text content a hard but important problem. Although text legibility has been studied in the context of style and linguistics [5], in this paper, we study it in the context of vision perception. For the rest of the paper, whenever we mention text quality or legibility, we refer to text image quality or legibility.

Although text legibility might be one of the most influential factors affecting visual ratings of text quality, they are not equivalent. Legibility is essentially a recognition task, i.e. whether the text is recognizable or not. Apart from the perceptual quality of the text, legibility perception is also influenced by other factors such as font, size, handwriting, occlusion, etc. Predicting text legibility, being closely associated with text quality in general, is not only important for media consumption but also can have a potential impact on text detection and recognition-based applications like image search [6], intelligent inspection [7], robot navigation [8], instant translation [9], industrial automation [10], etc. Scene text recognition is thus a challenging and widely studied problem in computer vision [1], but it also remains challenging due to texts appearing on complex backgrounds, in various fonts, and under imperfect imaging conditions [11]. Thus determin-



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ing the legibility of texts could assist in OCR applications.

Text legibility in the context of visual data has been only rarely studied in the literature and the goal of our work is to create a base for future research in text quality in online media content. COCO-Text [1] has been one of the pioneering datasets in scene text recognition. The creators sampled images with text from natural scenes and created a diverse dataset containing both machine-printed and handwritten texts. It was the only dataset we could find with human-annotated labels for text legibility, although most of the dataset is focused on embedded text. The authors did not study the effect of legibility on quality.

Image and video quality assessment has been studied extensively for UGC media with large open-source datasets available [12, 13, 14]. Although the models performed well while predicting the quality of natural scenes, we found they did not perform well on text content (Sec. 5), which may be due to the lack of such content in the training datasets or the significance of texts (amount of area it covers) in the UGC images. Regardless, the content of those datasets needs to be updated to account for the overwhelming popularity of short-form media. Towards progress in this direction, we make the following contributions:

- We modify the COCO-Text dataset to create a text legibility dataset consisting of $35K$ legible and $39.5K$ illegible text patches (Sec 3.1). Using the dataset we train deep text legibility predictors which perform very well on natural text images (Sec 4).
- Using the best-performing legibility predictor, which we call LegiT, we also study the effect of legibility on text quality and discuss how legibility features can be used to help build accurate text quality predictors (Sec 5). To mitigate the absence of text quality datasets in literature, we also introduce the LIVE-YT Text Quality dataset, which we describe briefly (Sec 3.2).

## 2. RELATED WORK

### 2.1. Text Legibility

Text legibility has been studied under different contexts - augmented/virtual reality, enhancing legibility of handwritten and generated texts, assisting people with low vision, and natural language processing (NLP). The authors of [15] studied the readability and legibility of texts displayed on different AR glasses and how environmental factors such as ambient luminance background properties, opacity, size, etc. affect the perception of legibility. In [16], the authors experimented with the influence of resolution and text complexity factors on the legibility of Chinese characters in virtual environments. In [5], the authors investigated the effects of font size, spacing, and margins in text interfaces that impact the digital reading of the elderly, whereas the authors of [17] showed how a computer graphic based adjustable typography can help visually-impaired users customize fonts to maximize legibility.

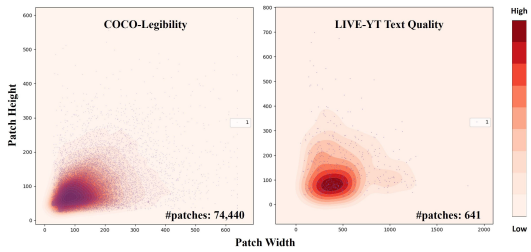


**Fig. 2**: **Distribution of resolutions:** Overlaid contour plot of text patch sizes (width vs height) in modified COCO-Text (left) and LIVE-YT Text Quality (right) datasets, where darker shade implies higher density of patches.

In the field of text enhancement, the authors of [18] proposed methods to enhance the legibility of ancient scripts, whereas [19] proposes a novel recurrent neural network for the alignment of handwritten texts to improve legibility. A novel scene text image super-resolution method was proposed in [20] using a Diffusion-based Prior-Enhanced Attention Network (PEAN) to enhance legibility for increasing recognition performance. Legibility has also been studied in the context of generated texts [21] in NLP, as well as generated word-as-image for semantic typography [22].

In our research, the COCO-Text dataset [1] was the only work we executed where the authors tried to study and collect legibility labels for natural scene texts. Along with transcription and category labels, they collected binary legibility annotations for scene texts in images taken from MS COCO dataset [23]. However, they did not propose a legibility predictor. We fill this gap and also model text quality as well.

### 2.2. No-reference Image Quality Assessment

No-reference Image Quality Assessment (NR-IQA), crucial in industry and computer vision, has garnered significant research attention. Highly successfull methods like BRISQUE [24] and NIQE [25] extract features with regression models [24, 25, 26, 27, 28], whereas codebook-based approaches like CORNIA [29] and HOSA [30] use visual codebooks for quality-aware features. Recent supervised methods have made a lot of progress [31, 32, 33, 12, 34]. Hyper-IQA [32] employs a self-adaptive hypernetwork, while Su et al. [33] adopts a supervised approach to learn the distortion manifold.

Supervised methods suffer due to scarce labeled data, which is costly and time-consuming to collect. Self-supervised learning emerges as a promising NR-IQA technique [35, 36, 37], training an encoder on unlabeled data with a contrastive loss. SPIQ [37] introduces a quality-aware contrastive loss, CONTRIQUE [35] treats representation learning as a classification task, and Re-IQA [36] trains quality and content-aware encoders. Existing methods aim to maximize the similarity between representations of crops from the same distorted



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image. Due to the dearth of data, both in the context of legibility and perceptual quality for UGC textual content, we apply transfer learning to generate legibility-aware features, which when combined with quality-aware features gathered from the state-of-the-art IQA models, leads to satisfactory prediction of text quality.

**Fig. 3**: **Extracting text patches:** Sample frames and corresponding text patches extracted from them as present in the LIVE-YT Text Quality Dataset.

## 3. DATASET

### 3.1. Text Legibility Dataset

#### *3.1.1. COCO-Text Dataset:*

The COCO-Text dataset [1] consists of images containing scene text sampled from the Microsoft COCO dataset [23]. The dataset is diverse with texts ranging from machine-printed street signs to illegible graffiti and handwritten notes. COCO-Text aimed to advance text detection and recognition algorithms for real-world contexts. It consists of $63,686$ images with $173,589$ labeled text regions, with each text region labeled by a bounding box, classification in terms of legibility, category (machine printed vs handwritten), script of the text (English or not), and transcriptions for legible texts with western script.

#### *3.1.2. Creating Legibility Dataset:*

From their study data analysis, the authors of COCO-Text [1] state a human benchmark of $87\%$ as the accuracy for determining legibility. To study text legibility, we built our own modified text patch dataset. For each image containing text annotations, we ignored all other classifications (category, language, transcription, etc) except for legibility. This dataset has very small size patches, with overlapping patches of similar legibility. The goal was to extract patches that would have a significant effect on the quality perception of the entire scene, similar to the text quality dataset we describe later, hence we merged small patches that had overlapping bounding boxes or that were separated by a margin of 20 pixels or less. An example of merging patches is shown in Fig. 1. The legibility label was chosen based on the majority area. As a result, we ended up with $74,440$ text patches ($34,927$ legible and $39,513$ illegible) from $173.6K$ text regions. We preserved the original train-validation split of COCO-Text, resulting in $60.4K$ training images and $14K$ validation patches. The variation of patch sizes in the created dataset is shown in Fig. 2. Sample legible and illegible patches are shown in Fig. 4.

**Fig. 4**: **Examples of text patches from the two datasets:** Samples of legible and illegible text patches from the modified COCO-Text dataset (top), and high and low quality text patches from the LIVE-YT-TQ dataset (bottom).

### 3.2. Text Quality Dataset

Because of the absence of a UGC image dataset targeting text quality, we conducted a human study whereby we collected subjective labels of text quality. We refere to this new resource as the LIVE-YouTube Text Quality (LIVE-YT-TQ) Database.

#### *3.2.1. Dataset Description*

We sampled a total of $405$ videos from the Internet Archive, each cropped to an approximate duration of 7 seconds. While sampling, we used Amazon Rekognition [38] to detect videos containing text in them. We mostly focused on videos with overlaid texts with a few of them containing in-scene embedded text. The video content was diverse consisting of vlogs, news, gaming, etc. From the video dataset, we cropped $641$ text patches, ignoring patches with less than $4\%$ of the frame area. Fig. 3 shows samples of frames and corresponding text patches extracted from them. Sample high and low quality patches are shown in Fig. 4 (bottom).

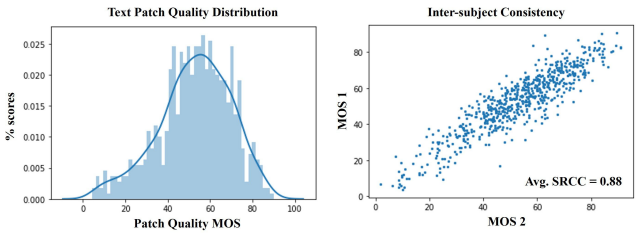


**Fig. 5**: **Left:** Distribution of mean opinion scores of the subjective quality labels for the text patches in the LIVE-YT Text Quality dataset. **Right:** Scatterplot showing the high correlation between the calculated MOS of two random splits of subject annotations. A high value of average SRCC indicates high consistency of the subjective data.

#### *3.2.2. Human Study*

We conducted an in-lab subjective quality study where participants provided perceptual quality ratings on text patches displayed on an FHD laptop screen. Although they were asked to rate the patch quality as a whole, they were informed the



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objective of the study was to assess the quality of text in them. Each subject was asked to rate 290 patches per session. The text patches were presented at their native resolution and not upscaled to prevent any loss of quality due to resizing. 28 volunteer subjects participated in the study providing around 18 ratings per text patch. The MOS (mean opinion score) distribution of text patch quality is shown in Fig. 5 (left). We also divided the subjects into two random groups and calculated the correlation of MOS between the two groups. This experiment was carried out for 50 random splits and the observed correlations were averaged, obtaining a mean inter-subject consistency of SRCC 0.88 (Spearman Rank Correlation) (Fig. 5 (right)), which validates the quality of our data.

## 4. MODELING LEGIBILITY

We use the dataset described in Sec 3.1 to build algorithms that predict the legibility of text patches. This set of models can then be further used for text quality prediction and also as a preprocessing step for text recognition and enhancement applications. The following section describes the performance of some deep backbones in legibility prediction. As the target variable is binary, we use a threshold of 0.5 to determine the prediction accuracy.

### 4.1. Preprocessing and Training Parameters

The training dataset consists of $60,435$ text patches – $28,367$ legible and $32,068$ illegible. The test dataset consists of $14,005$ text patches – $6,560$ legible and $7,445$ illegible. This split is kept consistent with COCO-Text [1]. The $60.4K$ training patches were further split randomly into $85/15$ training-validation sets resulting in $51.4K$ training and $9K$ validation data. When training, all the patches were center cropped and padded to a size of $128 \times 128$. We also applied random rotation of factor 0.1 (10% of $2\pi$) and random zoom of factor 0.1 as data augmentation techniques. We used a batch size of 128, and the training was done in two phases for stability – with the backbone layers frozen, the head was trained for 10 epochs at a $lr = 10^{-3}$ and then some of the backbone layers were fine-tuned for 20 epochs at $lr = 10^{-4}$.

### 4.2. Results

We compare the performance of ten popular backbones pre-trained on ImageNet [48]. The results are shown in Table 1. Along with the test prediction accuracy before and after fine-tuning, we also state the number of features pooled from the backbones, the number of layers unfroze while fine-tuning them, the size of the head layers, and the size of the backbones. We initially experimented with the same number of head layers, but for backbones with a larger number of output features, the number of trainable parameters in the head became very high compared to the dataset size. So, for those backbones, we reduced the head size to prevent overfitting and get reasonable performance. The model with MobileNetV2 [47] obtained the best accuracy (81.21% on test) when it was fine-tuned end to end. All of the other models were too deep and hence overfitted with further training. Although this list is not exhaustive, we believe the trained models are a good base to study and build text-quality models. We refer to the top-performing model (MobileNetV2 backbone) as LegiT, which is employed to extract legibility features for the text quality prediction models as described in Sec. 5.

## 5. LEGIBILITY TO QUALITY

### 5.1. Testing UGC IQA models

While developing a text quality prediction model, we first tested the performance of state-of-the-art UGC image quality models on the LIVE-YT-TQ dataset (Sec 3.2). Among the popular SOTA models, we evaluated the following without fine-tuning: 1) PaQ-2-PiQ RoIPool [12] pre-trained on FB-LIVE, 2) MUSIQ [31] pre-trained on FB-LIVE KonIQ-10K [49], and SPAQ [50], and 3) KonIQ++ multi-task model [51]. Although these models perform well on UGC images, their performance on UGC text images were subpar: PaQ-2-PiQ (SRCC 0.43, PLCC 0.45), MUSIQ (SRCC 0.40, PLCC 0.41), and KonIQ++ multi-task model (SRCC 0.56, PLCC 0.54); where SRCC is the spearman rank correlation coefficient and PLCC is the Pearson correlation coefficient.

### 5.2. Training on deep features

The goal of the following experiments was to observe how introducing legibility features can improve text quality prediction. Because our text quality dataset is relatively small (641 patches), fine-tuning a deep model led to overfitting. Hence, we trained a Support Vector Regressor (SVR) on features extracted from the backbones. For feature extraction, the patches were zero-padded to $512 \times 512$, preserving size and aspect ratio. The output of the backbones was passed through a MaxPooling layer to obtain 2048 features, which was larger than our dataset size and the feature set was sparse (40 – 70%). Hence we reduced the feature set to 50 components using Principal Component Analysis (PCA), which were subsequently used to train the SVR. For training, we used an $80:20$ train-test split with results averaged over 100 random splits.

### 5.3. Results

We notice that the backbones (ResNet50V2 [41], Xception [43], and ConvNeXt-T [40]) trained on ImageNet [48] performed a bit better than the previously discussed non-finetuned UGC IQA models. The IQA models (PaQ-2-PiQ [12], MANIQA [52], and MUSIQ [31]) performed better than the aforementioned deep backbones as they were trained on



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| Model Backbone | #Features | Head Layers | #FT Layers | Acc. before FT | Acc. after FT | #Parameters |
|---|---|---|---|---|---|---|
| EfficientNetV2S [39] | 1280 | 128 + 128 + 32 | 32 | 72.37 | 75.19 | 21.6M |
| ConvNeXtSmall [40] | 768 | 256 + 128 + 32 | 5 | 75.77 | 76.04 | 50.2M |
| ConvNeXtBase [40] | 1024 | 128 + 128 + 32 | 5 | 73.92 | 75.15 | 88.5M |
| ResNet50V2 [41] | 2048 | 128 + 32 | 4 | 78.78 | 78.94 | 25.6M |
| ResNet152V2 [42] | 2048 | 128 + 32 | 4 | 78.83 | 79.91 | 60.4M |
| Xception [43] | 2048 | 128 + 32 | 3 | 78.71 | 79.21 | 22.9M |
| InceptionV3 [44] | 2048 | 128 + 32 | 10 | 78.61 | 79.51 | 23.9M |
| InceptionResNetV2 [45] | 1536 | 128 + 128 + 32 | 4 | 78.18 | 79.96 | 55.9M |
| **DenseNet201** [46] | 1920 | 128 + 32 | 55 | **79.26** | 80.35 | 20.2M |
| **MobileNetV2** [47] | 1280 | 128 + 128 + 32 | all | 78.39 | **81.21** | 3.5M |
| **Human Benchmark [1]** | - | - | - | - | **87** | - |

**Table 1**: **Performance comparison of different backbones in legibility prediction** when trained on the modified COCO-Text dataset. Here FT stands for fine-tuned and performance is evaluated by percentage of accurate predictions on the test text patch dataset. The human benchmark stated in [1] has also been referenced. The top two performing models have been highlighted.

| Model | SRCC | PLCC | RMSE |
|---|---|---|---|
| ResNet50V2 | 0.60 | 0.65 | 12.93 |
| Xception | 0.66 | 0.69 | 12.06 |
| ConvNeXt-T | 0.69 | 0.71 | 11.81 |
| PaQ-2-PiQ | 0.69 | 0.70 | 12.07 |
| MANIQA | 0.71 | 0.72 | 11.97 |
| MUSIQ | 0.71 | 0.73 | 11.81 |
| **PaQ-2-PiQ + LegiT** | 0.72 | 0.75 | 11.12 |
| **MANIQA + LegiT** | 0.73 | 0.74 | 11.50 |
| **MUSIQ + LegiT** | **0.75** | **0.77** | **10.18** |

**Table 2**: **Performance of text quality models** when trained and tested on LIVE-YT Text Quality dataset. Higher values (SRCC and PLCC) indicate better performance. We notice that introducing legibility features improves the performance of SVRs.

image quality datasets leading to quality-aware features. For MUSIQ, quality features were extracted from the multi-scale encoder and for MANIQA, features were extracted from the Scale Swin Transformer Block. For the last three models, the legibility features were extracted from LegiT (the best-performing legibility prediction model with MobileNetV2 backbone), as described in Sec 4, then appending to the quality features and finally pooling, led to a $3-6\%$ gain in performance. These results are summarized in Table 2.

## 6. CONCLUSION

Assessing text legibility and quality is a very important problem for evaluating UGC media, given textual content is so prevalent across the internet. Moreover, assessing legibility could also help streamline other text recognition-based applications. Overlaid text is also an important part of the content in professional media categories such as sports, news, movie subtitles, etc. Even with this scale of prevalence, very little has been explored in the vision literature. We develop a text legibility prediction model based on a modified version of COCO-Text and also study its impact on creating accurate text quality predictors. For future work, we want to develop more advanced methods that combine text legibility and text quality. We also introduce a first-of-its-kind text quality dataset for UGC media. We hope this work will be a strong base for further research of media content in perceptual quality evaluation and we look forward to expanding this in the near future.

## 7. ACKNOWLEDGMENTS

This research was sponsored by a grant from YouTube, and by grant number 2019844 from the National Science Foundation AI Institute for Foundations of Machine Learning (IFML). The authors thank The University of Texas at Austin for providing computational resources that have contributed to the research results reported in this paper.

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